

Predicting Maternal Mortality Rates Using Socioeconomic, Demographic, and Health Infrastructure Indicators

**Capstone Project Proposal Submitted in Partial Fulfillment of the Requirements for the
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by

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1 Project Description

Maternal mortality remains one of the most persistent global health challenges despite advances in medicine and public health infrastructure. Significant disparities exist across income levels, racial groups, geographic regions, and environmental conditions. In this project, I design an interpretable, reproducible data science framework to analyze and forecast maternal mortality ratios (MMR) using socioeconomic, demographic, healthcare, and environmental indicators at the country-year level.

Despite technological and clinical improvements, global maternal mortality reductions have slowed in recent years, and many regions remain far above the Sustainable Development Goal (SDG) target of fewer than 70 deaths per 100,000 live births by 2030. Policymakers and health researchers lack accessible analytical tools that integrate economic development, healthcare access, fertility dynamics, and environmental stressors into predictive models. I address this gap by building a structured machine learning pipeline that quantifies how these factors contribute to maternal mortality and projects future trends under data-driven assumptions.

The objective of this project is to integrate global maternal mortality data with World Bank socioeconomic and environmental indicators, conduct comprehensive exploratory data analysis to uncover structural patterns, build and compare predictive models including linear, ensemble, kernel-based, and neural network approaches, and provide interpretable insights using feature importance, permutation importance, and partial dependence analysis. The outcome will be a reproducible GitHub repository, analytical report, and forecasting framework that connects empirical findings to global health targets. Figure 1 presents historical global average maternal mortality trends and illustrates the persistent gap between current levels and SDG targets.

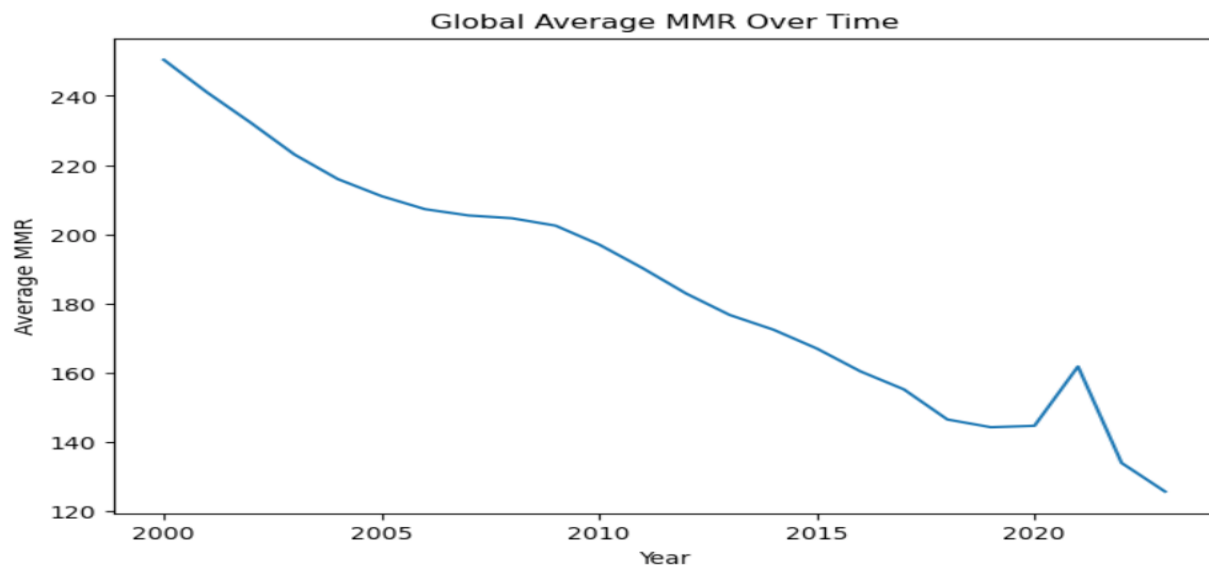


Figure 1 Global average maternal mortality ratio (2000–2023).

2 Literature Review

Extensive public health research identifies structural inequality as a primary determinant of maternal mortality. The World Health Organization (2023) documents strong associations between income level, skilled birth attendance, and maternal outcomes. Countries with limited healthcare infrastructure consistently experience higher maternal mortality ratios. In the United States, Centers for Disease Control and Prevention (2024) reports show persistent racial disparities, underscoring the role of structural inequities in shaping maternal health outcomes.

Global Burden of Disease studies conducted by the Institute for Health Metrics and Evaluation demonstrate the value of statistical modeling in estimating maternal mortality across countries and time. These studies validate the use of longitudinal panel data and predictive modeling approaches for maternal health analysis. Recent machine learning literature in public health shows that ensemble models such as Random Forest and Gradient Boosting frequently outperform traditional regression models when nonlinear relationships and interaction effects are present.

Environmental health research further links climate stressors and air pollution exposure to adverse pregnancy outcomes. Elevated PM2.5 exposure and extreme heat conditions are associated with increased risk of hypertensive disorders, preterm birth, and maternal complications. However, few studies integrate socioeconomic, healthcare, demographic, and environmental indicators within a unified predictive framework for global maternal mortality.

Additionally, many predictive studies lack interpretability, limiting policy relevance. This project addresses these gaps by combining diverse structural indicators within an interpretable machine learning pipeline and explicitly connecting model findings to global development targets.

3 Approach and Methodology

3.1 Research Design

This study follows a quantitative, observational, country-year panel design. The unit of analysis is the country-year observation. The dataset includes over 5,000 country-year records from 2000 to the most recent available year. All data are publicly available and aggregated at the national level.

3.2 Data Collection and Integration

I integrate maternal mortality data from global health agencies with socioeconomic and environmental indicators from the World Bank World Development Indicators database. Variables include GDP per capita, fertility rate, health expenditure as a percentage of GDP, skilled birth attendance, PM2.5 air pollution exposure, and climate stress measures such as heat index and cooling degree days. Data are merged using standardized ISO3 country codes and year identifiers.

3.3 Data Preprocessing

I assess missingness, ensure consistent variable definitions, and standardize measurement units. Because maternal mortality exhibits strong right skewness, I apply a log transformation to stabilize variance and improve modeling performance. Predictors are standardized using StandardScaler for models sensitive to feature magnitude.

3.4 Exploratory Data Analysis

I conduct comprehensive exploratory data analysis to understand distributional properties, time trends, and structural relationships. Correlation matrices quantify linear associations between predictors and maternal mortality, as shown in Figure 2. Scatter plots and boxplots illustrate socioeconomic gradients in maternal mortality. Time-series analysis evaluates trends and variability across years.

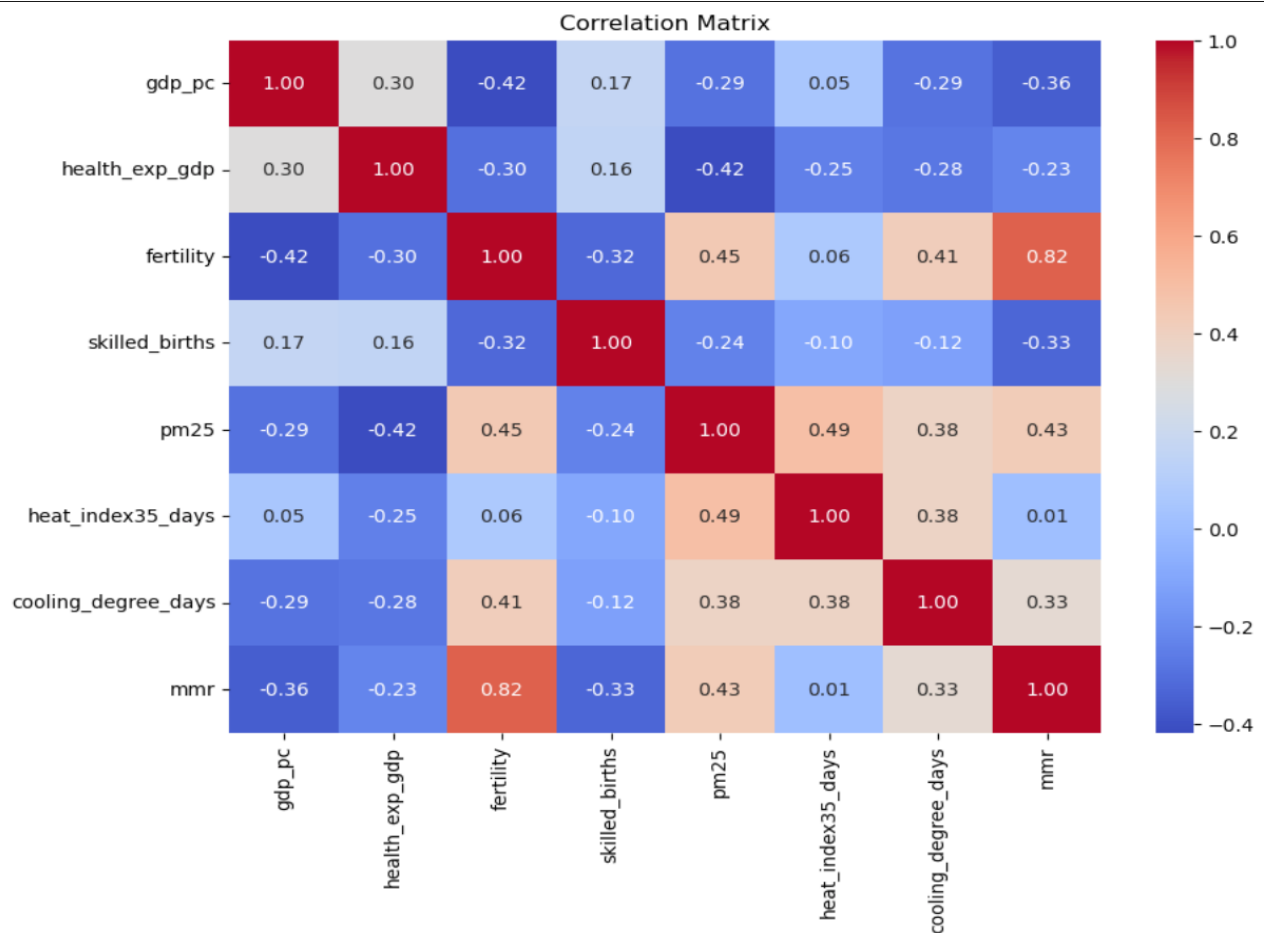


Figure 2. Pearson correlation matrix of predictors and maternal mortality ratio.

Exploratory findings indicate that fertility rate exhibits the strongest positive association with maternal mortality, while GDP per capita demonstrates a negative association consistent with development theory.

3.5 Modeling Strategy

I implement and compare multiple model families, including Linear Regression, Ridge Regression, Lasso Regression, Random Forest, Gradient Boosting, Support Vector Regression (RBF kernel), and Multilayer Perceptron (MLP) neural networks. I apply a time-based train-test split to preserve temporal integrity, training models on years up to 2018 and evaluating performance on subsequent years.

I evaluate model performance using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 on the log-transformed scale. Figure 3 presents predicted versus actual maternal mortality values for the best-performing model.

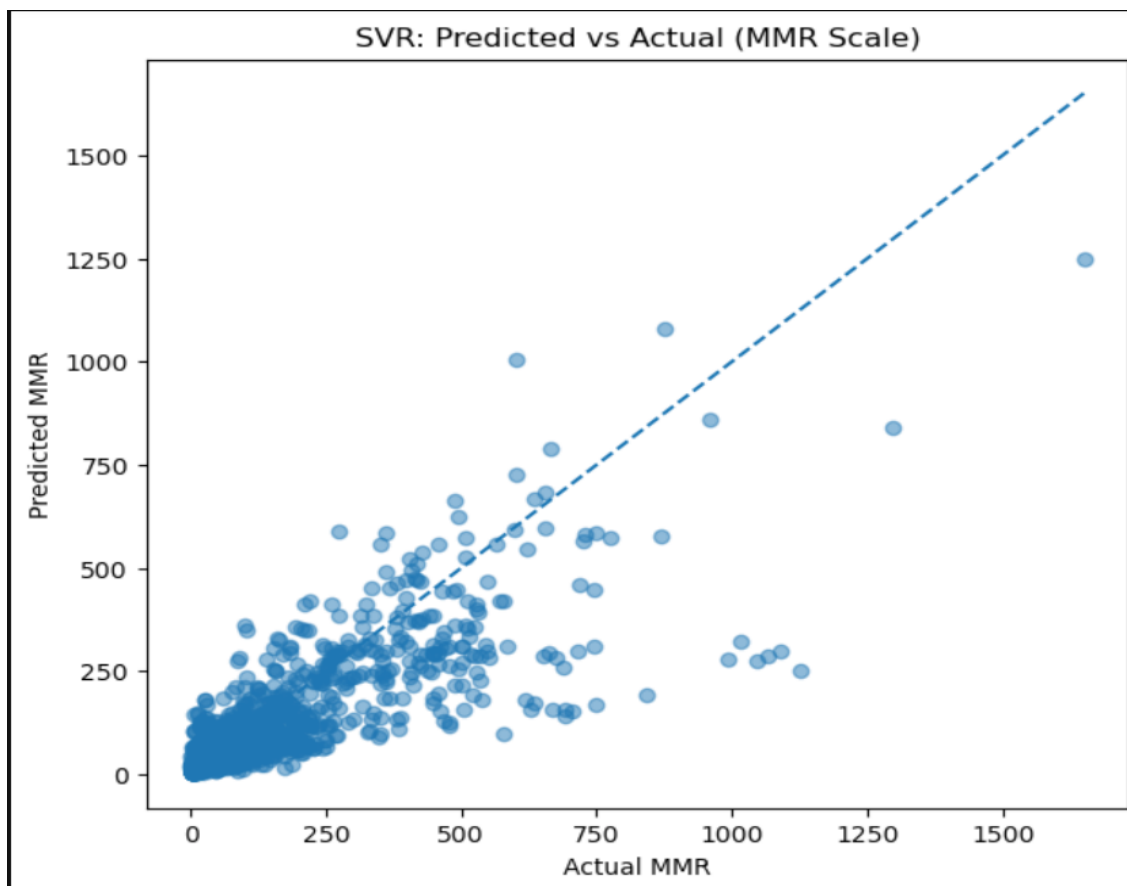


Figure 3. Predicted versus actual maternal mortality ratio with the best performing model (SVR).

3.6 Model Interpretation

To ensure interpretability, I compute permutation feature importance and generate partial dependence plots. Permutation importance measures how model performance changes when individual features are randomly shuffled. Results indicate that GDP per capita and fertility rate are the most influential predictors. Figure 4 displays the permutation importance ranking.

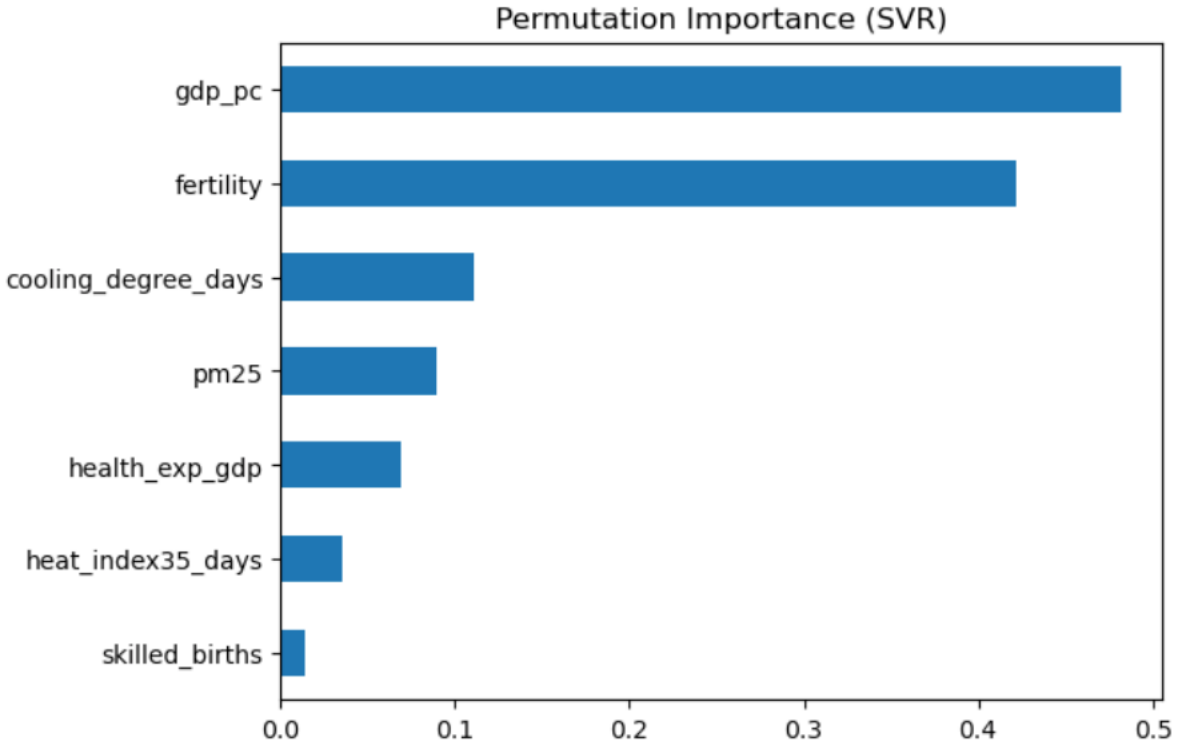


Figure 4. Permutation feature importance ranking.

3.7 Forecasting Framework

Using the best-performing model, I generate projections of maternal mortality through 2030. Predictor variables are extrapolated using historical country-specific trends. The model then estimates future maternal mortality ratios under a baseline continuation scenario. Figure 5 illustrates projected global maternal mortality trends relative to the SDG target.

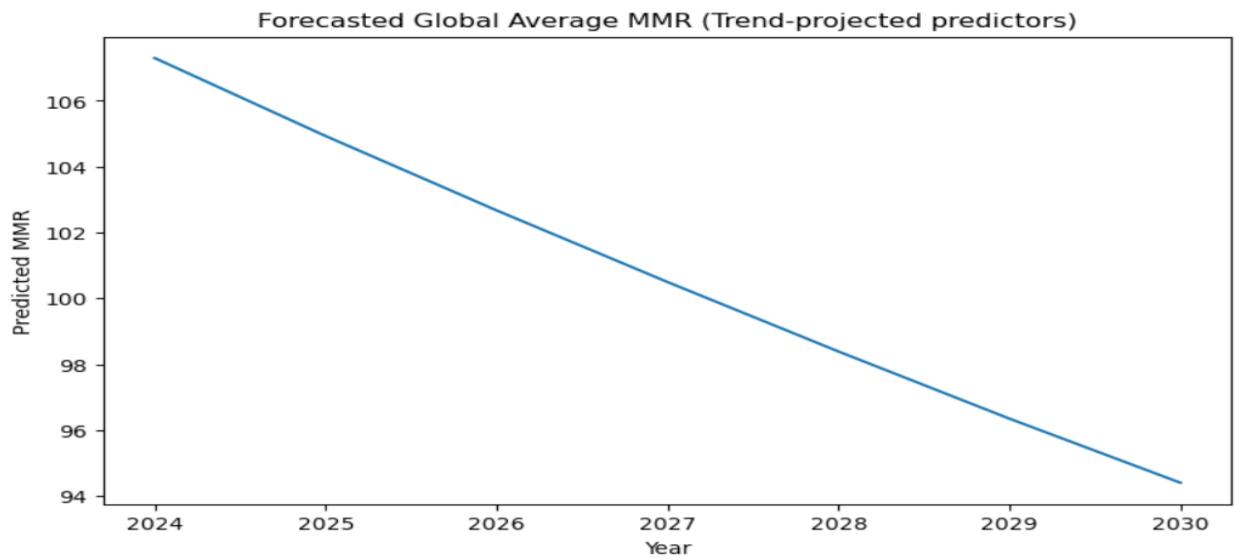


Figure 5. Project maternal mortality ratio through 2030.

This forecasting approach provides a data-driven baseline estimate of whether current structural trends are sufficient to achieve SDG targets.

3.8 Limitations

This study relies on publicly available aggregate data, which may contain reporting inconsistencies. Climate variables may vary in measurement precision across regions. Forecasting assumes continuation of past trends and does not account for sudden policy shifts or global shocks.

4 Deliverables

The final deliverables include a cleaned and integrated country-year dataset, a comprehensive exploratory analysis notebook, multiple predictive models with performance comparison, an interpretable feature importance and model explanation report, a forecasting framework through year 2030, a reproducible GitHub repository, a formal written capstone report, and presentation slides for the project defense.

5 Technology

This project uses Python with Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, SHAP (where applicable), and Jupyter Notebooks. Version control is maintained using GitHub. Climate and socioeconomic data are obtained from publicly available APIs and CSV downloads.

6 Dependencies and Limitations

Project completion depends on consistent availability of World Bank and global health datasets. Differences in national reporting standards may introduce measurement variability. Forecasting assumes stability in predictor trends.

6.1 Ethical/Privacy/Cybersecurity Concerns

All data is publicly available, de-identified, and aggregated at the country level. No personally identifiable information is used. Data storage and code repositories comply with institutional standards for data security and academic integrity. Proper citation and data use compliance will be maintained.

7 Project Timeline

Milestone	Target Date
Data collection & merging	September 2025
EDA & statistical analysis	October 2025
Modeling (ML + Deep Learning)	January 2026
Model evaluation & interpretation	February 2026
Draft report completion	March 2026
Revisions & final results	April 2026
Final project report submitted	April 2026
Oral presentation	April 2026

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